Validity Analysis: MMLU

Target context: Non-Arab Tourists and Expats in Eight Arab Countries

Overall risk: **HIGH**

Deployment Context

Use case: Support foreigners visiting Morocco, Egypt, Jordan, Palestine, Lebanon, UAE, Kuwait, or KSA in answering their general questions

Target population: Non-Arab tourists or expats visiting Morocco, Egypt, Jordan, Palestine, Lebanon, UAE, Kuwait, or KSA and have general questions about the Arabic language, history or geography of these countries

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ⚠	1	Serious concern	high
Input Content ⚠	1	Serious concern	high
Input Form ✓	3	Moderate gaps	high
Output Ontology ⚠	1	Serious concern	high
Output Content ⚠	1	Serious concern	high
Output Form ✓	2	Significant gaps	high
Average	**1.5**		

⚠ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

MMLU is fundamentally misaligned with this deployment across the four HIGH-priority dimensions (IO, IC, OO, OC). Documentation, web evidence, and dataset analysis converge on a clear finding: MMLU's 57 subjects are anchored in US academic curricula with no dedicated Arab-region content; ground-truth labels reflect US-academic and (in moral_scenarios) explicitly US-2020 framings; and the single-answer MCQ schema is structurally incapable of representing the multi-perspective output behavior required for contested Arab historical and political topics. The two MODERATE-priority dimensions (IF, OF) match deployment modality at the surface (English text) but cannot evaluate the Arabic-learning sub-cohort or open-ended explanatory output. Universal STEM and formal-reasoning content remains usable as a transferable competence baseline, but the core deployment domain — Arab history, regional geography, Arabic language basics, and Islamic religious knowledge — is essentially uncovered.

Practical Guidance

What This Benchmark Measures

MMLU can usefully measure transferable academic competence in subjects with no cultural dependence — STEM, formal logic, mathematics — and can serve as a baseline for English-language reading comprehension and MCQ reasoning. It cannot measure the deployment's core constructs: Arab history knowledge, Middle Eastern geography, Arabic language basics, Islamic religious knowledge, or the ability to acknowledge multiple perspectives on contested topics.

Construct Depth

Construct depth is shallow for the deployment's purpose. The benchmark probes broad academic recall via MCQ but cannot surface (a) regional factual accuracy on Arab content, (b) framing perspective on contested events, (c) calibrated hedging on politically sensitive topics, or (d) Arabic-language competence. Even the calibration evaluation [Q64–Q67], while informative about confidence-accuracy gaps, cannot diagnose region-specific overconfidence patterns documented in CAMeL and MENAValues research [WEB-11, WEB-15].

What Else You Need

Substantial supplementation is required. For IO and IC: ArabicMMLU [WEB-2] (40 tasks, 8 countries, 14,575 MSA questions) and ILMAAM [WEB-5] (culturally refined with 11-expert annotation). For dialect/IF coverage of the Arabic-learning cohort: DialectalArabicMMLU [WEB-1] (Moroccan, Egyptian, Emirati, Saudi). For OC value alignment across all eight countries: MENAValues [WEB-11]. For Islamic-cultural content: PalmX 2025 [WEB-13] and IslamicMMLU [WEB-14]. None of these supplements solve the OO contested-topic schema gap, which is a confirmed ecosystem-wide gap (FV5) and would require new evaluation methodology (multi-perspective scoring, expert adjudication of contested labels).

ID	Evidence
Q64	“We should not trust a model's prediction unless the model is calibrated, meaning that its confidence is a good estimate of the actual probability the prediction is correct.” (p.7)
Q65	“We evaluate the calibration of GPT-3 by testing how well its average confidence estimates its actual accuracy for each subject.” (p.7)
Q66	“Its confidence is only weakly related to its actual accuracy in the zero-shot setting, with the difference between its accuracy and confidence reaching up to 24% for some subjects.” (p.7)
Q67	“Another calibration measure is the Root Mean Squared (RMS) calibration error (Hendrycks et al., 2019a; Kumar et al., 2019).” (p.7)
WEB-1	DialectalArabicMMLU covers five Arabic dialects (Syrian, Egyptian, Emirati, Saudi, Moroccan) with substantial cross-dialect performance variation (https://arxiv.org/abs/2510.27543)
WEB-2	ArabicMMLU is the closest available substitute for the Arabic-language secondary cohort (14,575 MSA questions) (https://aclanthology.org/2024.findings-acl.334/)
WEB-5	ILMAAM (2025) identified significant cultural misalignments in religion and morality content; 11 expert annotators reviewed for cultural appropriateness, bias, religious sensitivity (https://aclanthology.org/2025.loreslm-1.29/)
WEB-11	MENAValues evaluates LLMs across neutral, personalized, and observer framings — illustrating richer output-form evaluation that MMLU's MCQ accuracy cannot match (https://arxiv.org/html/2510.13154)
WEB-13	https://arxiv.org/html/2509.02550
WEB-14	https://arxiv.org/html/2603.23750
WEB-15	Naous et al. 2024 (CAMeL) empirically demonstrate LLM Western cultural bias in Arabic prompts (https://venturebeat.com/ai/large-language-models-exhibit-significant-western-cultural-bias-study-finds)